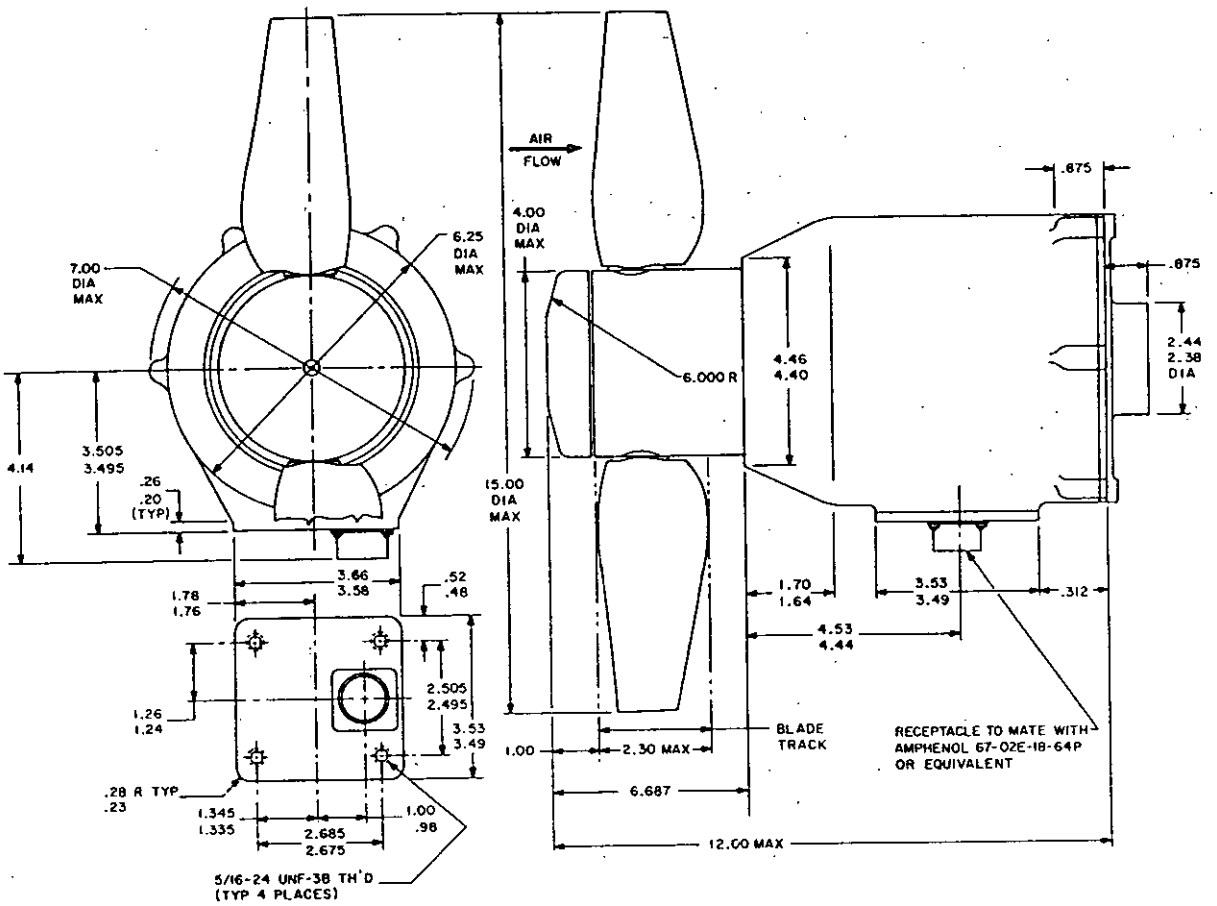
MILITARY SPECIFICATION

GENERATOR SYSTEM, EMERGENCY, 2.5 KVA, 400 HERTZ, ALTERNATING CURRENT, RAM AIR TURBINE DRIVEN, DETAIL SPECIFICATION FOR

The complete requirement for procuring the generating system described herein shall consist of this document and the issue in effect of MIL-G-21480, as amended below.

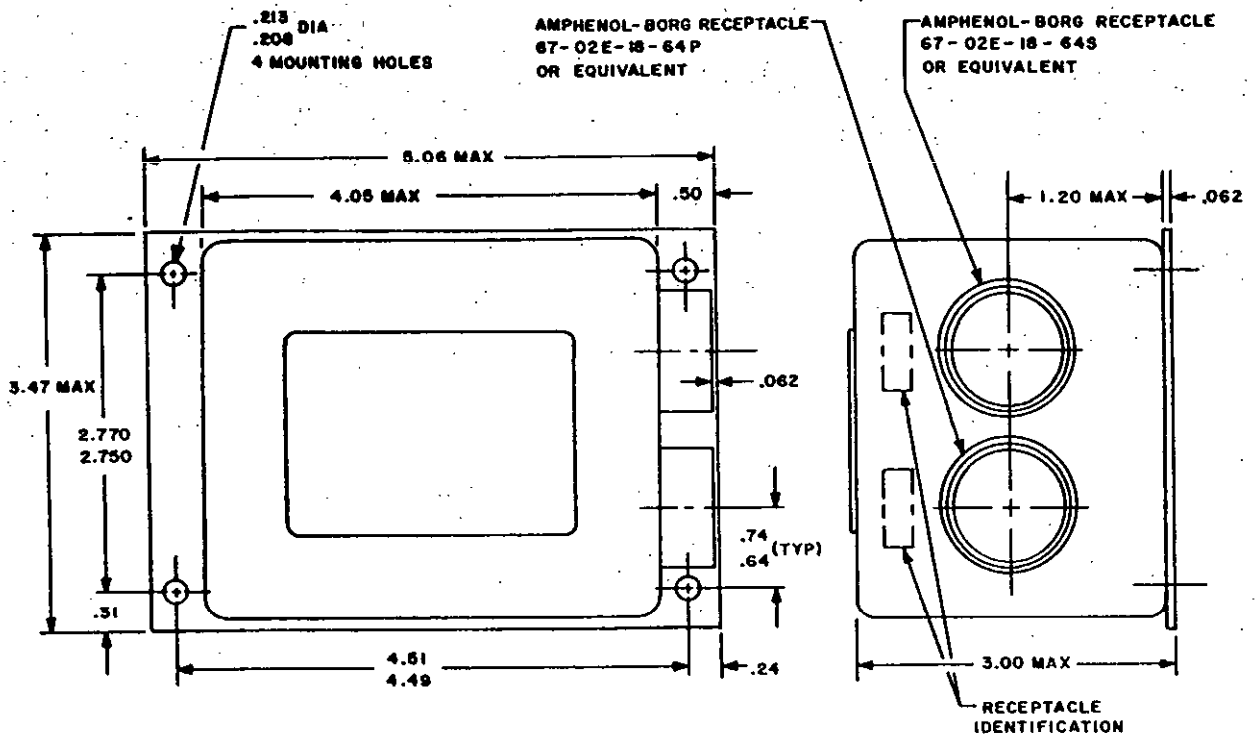


RAM AIR TURBINE GENERATOR - MAXIMUM WEIGHT 27 LBS
P/N M21480/3-1

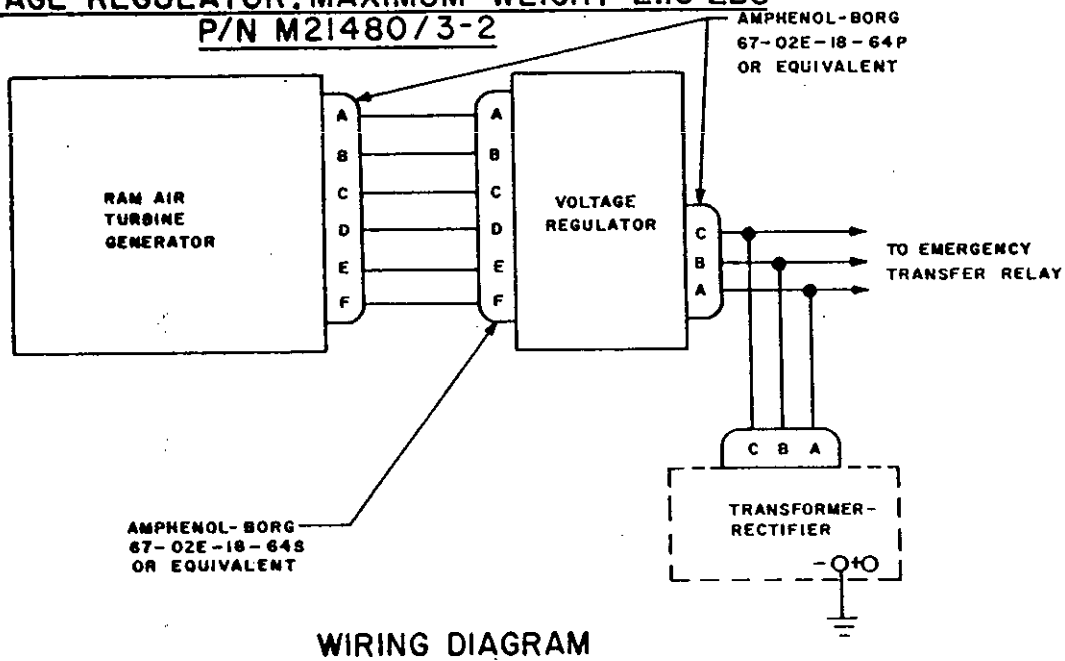
EMERGENCY GENERATOR SYSTEM DATA	
RATED LOAD:	2.5 KVA AT POWER FACTORS (pf) OF FROM .85 LAGGING TO 1.0 OR 2.0 KVA AT A pf OF .75 LAGGING WHEN THE TURBINE IS SUBJECTED TO A FREE AIR STREAM OF 100 KIAS TO MACH I INTO THE TURBINE PARALLEL TO THE AXIS OF ROTATION AND .5 KVA AT A pf OF .75 LAGGING TO 1.0 AT AIR SPEEDS OF 70 TO 110 KIAS.
VOLTAGE:	3-PHASE 115 ± 5 VOLTS LINE-TO-NEUTRAL
FREQUENCY:	400 ± 20 Hz
ALTITUDE:	SEA LEVEL TO 55,000 FT

FSC 6115

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VOLTAGE REGULATOR, MAXIMUM WEIGHT 2.15 LBS
P/N M21480/3-2



WIRING DIAGRAM

IDENTIFICATION OF ITEMS	
ITEM	MILITARY PART NUMBER
EMERGENCY GENERATOR SYSTEM (CONSISTS OF ONE RAM AIR TURBINE GENERATOR AND ONE VOLTAGE REGULATOR)	M21480/3-3
RAM AIR TURBINE GENERATOR	M21480/3-1
VOLTAGE REGULATOR	M21480/3-2

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1. The requirements of MIL-G-21480 pertain to the specification paragraphs as listed, with the following exceptions:

Delete all reference to "generator" and add "Ram Air Turbine Generator".

Delete all reference to supervisory panels, supervisory functions, and the requirements of 3.4.7, 3.4.10.1, 3.5.5, 3.5.7.1, 3.5.7.7, 4.4.1.1.1, 4.4.2.1.1, 4.4.2.2, 4.4.3.4, 4.5.6, 4.5.15, 4.5.19.

1.1 Scope. Add, "and ram air turbine".

3.4.8.2 Overload capacity. Change to read: "The system shall be capable of delivering rated load continuously. Also, the system shall deliver 150 percent rated current for 5 minutes at air speeds into the turbine of from 180 K to Mach I and 200 percent rated current for 5 seconds at air speeds into the turbine of 180K to Mach I without damage."

3.4.9.3 Electric connections. The Ram Air Turbine shall be supplied with a receptacle to mate with Amphenol 67-02E-18-64P or equivalent. The pin connections shall be as follows: A, B, and C shall be the three sequential phases; D shall be the single neutral; and E and F shall be the two ends of the shunt field windings.

3.4.9.6 Cooling. Change to read: "The generator shall be cooled by air flow from the turbine."

3.4.9.7 Change to read: "Life of Generator Turbine. The unit shall have a minimum life of 1000 hours to first overhaul during which it shall be capable of operating for 250 hours, minimum for intervals of 2 hours, minimum.

3.4.13 Nameplate data. The following information shall appear on the nameplate: "Name of Unit; Federal Stock Number, Military Part Number, Manufacturer's Name (no address); Manufacturer's Part Number; Contract Number; Government Acceptance Stamp, Acceptance Date and Serial Number.

3.5.1 Voltage limits. Change to read: "The generated phase voltage, nominally 115 volts rms, shall not exceed 120 nor be less than 110 volts rms over the airspeed and load range as specified herein for all steady state conditions. There shall be no residual voltage modulation in the system. Maximum 0.85 of lagging load applied and removed shall not cause the voltage to be greater than 120 volts or less than 110 volts, line-to-neutral, for longer than 0.2 seconds over the range of airspeeds specified."

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3.5.3 Phase Displacement and Voltage Imbalance. Change to read: "A 1/4-load imbalance shall not cause more than 6 percent voltage imbalance. A 1/5 imbalance shall not cause more than a 4 percent voltage imbalance."

3.5.4 Wave form. Delete "±5%" and add "±15%".

Delete "1.5%" and add "5%" for any single harmonic.

Delete "3%" and add "8%" for total harmonic.

3.5.7.2 Fault clearing. Change to read: "No fault of the system, including those defined in 6.2, shall cause the voltage of any phase of the generating system to exceed the limits of figure 6".

Add: 3.9 Frequency. The generated frequency, nominally 400 hertz, shall not exceed 420 nor be less than 380 hertz over the airspeed range as specified herein and from no load to rated load for all steady state conditions. There shall be no residual frequency modulation in the system. Maximum 0.85 power factor (pf) lagging load applied and removed shall not cause the frequency to be greater than 420 hertz or less than 380 hertz for longer than two seconds over the range of airspeeds specified for rated loads.

3.10 Acceleration. The system shall be capable of withstanding the following accelerations. The system must operate within prescribed limits during the accelerations shown by Column A, after those shown by column B, and shall maintain mechanical integrity but need not be operable after those shown by column C. All tests shall be run with mounting surface down. Accelerations specified in g-units.

Direction of Accelerator	A	B	C
(a) Along axis of rotation in the direction of air stream	1	5	40
(b) Along axis of rotation in direction opposite to (a)	1	10	40
(c) In a horizontal direction at 90 degrees to the axis of rotation. (both directions)	3	10	40
(d) Up	5	20	40
(e) Down	10	20	40

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4.4.1.2.1 Change to read: "Air Flow. The turbine generator shall be subjected to a free air stream of 110KIAS at a temperature of 25 ± 15°C".

4.4.3.2 Starting and warmup. Change to read. "With full load connected, the system shall perform within specified limits for operation within seven seconds after the system is dropped into a 110-knot air stream and in five seconds after the system is placed in a 180-knot air stream. No warmup shall be required for the proper operation of this system."

4.5.8.1 Qualification test. Delete all after (a) and add: "(a) Demonstrate that 1/4 load imbalance does not cause the voltage to be unbalanced by more than six percent. (b) Demonstrate that 1/5-load imbalance does not cause the voltage to be unbalanced by more than four percent."

4.5.14 Endurance. Change to read: A 100 hour (minimum) flight service test shall be conducted on an A-6 aircraft. A minimum of 10 actuations and a minimum of 20 operating hours during this period is required over the IAS and altitude operating range. At least two of the hours shall be during precipitation.

4.5.20 Salt fog. The Ram Air Turbine Generator shall be subjected to the salt fog tests, Procedure I, Method 509, MIL-STD-810B. The required salt solution shall be a synthetic sea water solution and shall be formulated as prescribed as follows:

Salt	Grams Per Liter
Sodium Chloride	24.540
Magnesium Chloride	11.110
Sodium Sulphate	4.094
Calcium Chloride	1.159
Potassium Chloride	0.695
Sodium Bicarbonate	0.201
Potassium Bromide	0.101
Strontium Chloride	0.042
Boric Acid	0.027
Sodium Fluoride	0.003

Adjust the pH of this solution, if necessary, to 8.2 with 0.1 normal sodium carbonate solution.

The test item shall be oriented so as to expose the most critical or vulnerable parts to the salt fog. The duration of the exposure period shall be 96 hours and the equipment shall be nonoperating. At the completion of this exposure period, the test item shall be dried for 24 hours. The test item shall then be operated and inspected for evidence of impairment. Metal parts shall be examined for corrosion, moving parts for clogging or binding, plastic parts for distortion, paint for blistering, and hygroscopic materials for swelling. Any deterioration or change in performance shall not exceed that defined in 4.5.34.

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4.5.23.1 Vibration. The Ram Air Turbine Generator shall be vibrated in accordance with MIL-STD-810B, Method 514.1, Procedure I. Procedure I shall be accomplished using vibration test curve AT of figure 514.1-1 and the test time schedule of Table 514.1-1, for equipment mounted without vibration isolators. During the test, the equipment shall not be operated.

Any deterioration or change in performance of the test item shall not exceed that defined in 4.5.34. At the conclusion of the test, the test item shall deliver rated load in an airstream of 110KIAS at a temperature of 25 $\pm$ 15°C. The unit shall remain capable fo sustained operation over the airspeed range of 110KIAS to Mach I. There shall be no evidence of cracking or rupture of any component or of the mounting assembly itself.

4.5.24 Shock. The Ram Air Turbine Generator shall be subjected to the shock tests of MIL-STD-810B, Method 516.1, Procedures I and III. Procedure I tests shall be accomplished using a shock pulse in accordance with figure 516.1-2, of amplitude 15g, and a time duration of 11 milliseconds. Three shocks in each direction shall be applied along three mutually perpendicular axes of the test item (total of 18 shocks). At the conclusion of the test, the test item shall be operated and inspected. Any deterioration or change in performance shall not exceed that defined in 4.5.34. Procedure III (Equipment Crash Safety) shall be accomplished using a shock pulse in accordance with figure 516.1-2, of amplitude 30g, and a time duration of 11 milliseconds. The shocks shall be applied to the test item in each direction along each of three mutually perpendicular axes (total of 12 shocks). There shall be no failure of the mounting attachment, and the test item shall remain in place and not create a hazard. However, bending and distortion shall be permitted.

4.5.34 Failure criteria. The unit is considered failed if during or after any test herein the unit will not meet specified performance limits, or if the maintenance of the unit is impeded or if the unit requires the replacement of components.

NOTES:

1. All dimensions are in inches. All tolerances are $\pm$.06.

Preparing Activity
Navy - AS
Project No. 6115-N453

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